

Does token order matter? A controlled, cross-species study of tokenized animal vocalizations

(author)

Master’s thesis spin-off — working draft, non-archival

Abstract

A growing line of work tokenizes animal vocalizations into discrete sequences and applies sequence models, assuming token *order* carries communicative information. We test this premise rigorously across two species — Egyptian fruit bats (*Rousettus aegyptiacus*; graded vocal system) and common marmosets (*Callithrix jacchus*; structured calls) — and three model families (a small BERT with masked-token pretraining, k -NN with Levenshtein distance, and an order-agnostic bag-of-tokens classifier), comparing classification on *real vs. within-vocalization shuffled* token order with cross-individual splits and confidence intervals. We find: (1) token-order effects are small in both species (Δ macro-F1 $\approx$ 0.01–0.03); the token *multiset* dominates classification; (2) **sequence length is a confound** prior work ignores — order is indistinguishable from shuffled at short sequences in both species; (3) pooled, marmosets show a small but significant order benefit while bats show none, explained by a clean mechanism: marmoset token sequences have far higher **sequential dependency** (bigram predictability 0.78 vs. 0.13–0.33 for bats), *not* higher token diversity (a hypothesis we refute). Constructively, a domain-matched self-supervised (SSL) tokenizer is the best discrete repertoire for vocalization-level tasks; a positive-pair ablation shows the contrastive invariance *causally* selects which task the tokens serve. We also flag a label-leakage pitfall for per-context tokenization. Our controls (shuffle ablation, length stratification, cross-individual splits, CIs, leakage checks) are absent from recent token-sequence work and temper its central premise.

1 Introduction

Computational bioacoustics increasingly borrows the discrete-token, sequence-model toolkit from NLP. The implicit assumption is that token *order* carries information — i.e. that animal communication has combinatorial syntax exploitable by sequence models. Recent work [1] tokenizes marmoset/dog calls and applies k -NN with Levenshtein distance, framing the goal as *leveraging sequential structure*, but typically (i) never isolates whether order helps (no shuffle control), (ii) reports no CIs, (iii) does not control sequence length, and (iv) does not study behavioral context or graded systems. We supply the missing controls and add a mechanistic explanation and a constructive tokenizer.

Contributions: (1) a shuffle-controlled order ablation across two species, three model families and three tasks,

with cross-individual splits and CIs; (2) identification of sequence length as a confound; (3) a mechanistic explanation (sequential dependency, not diversity) for a modest cross-species order contrast; (4) a domain-matched SSL tokenizer that is the best discrete repertoire, with a causal positive-pair ablation; (5) a per-context leakage caveat and evidence that silhouette mis-ranks tokenizers.

2 Data and methods

Corpora. Bats (Prat et al. 2017): 153k segments, 8 behavioral contexts, 41 individuals, 250 kHz. Marmosets (InfantMarmosetsVox): 72,921 calls, 11 call-types, 10 callers, 44.1 kHz. Sub-units: bat *segments* (mel 21×32) or *frames* (per-segment mel frames); marmoset mel-spectrogram *frames* per call. **Tokenizers.** k -means on mel/UMAP; per-context; agglomerative; and a domain-matched **SSL** encoder (NT-Xent; positives = two sub-units of the same vocalization; no labels) + k -means. **Order control.** tokens kept in temporal order (real) or permuted within the vocalization (shuffled); Δ = macro-F1(real) – macro-F1(shuffled). **Models.** TinyBERT (2 layers, $d=64$, MLM + [CLS]); k -NN + normalized Levenshtein; bag-of-tokens logistic regression; causal TinyBERT for next-token. **Protocol.** bats cross-individual (30/11), 5 seeds; marmosets stratified split, 5 seeds; macro-F1, 95% CIs over seeds.

3 Results

Order is (nearly) irrelevant for bats; small for marmosets. Fig. 1 shows Δ (real–shuffled) with CIs: all bat conditions straddle zero; both marmoset tasks are significant but small. bag $\approx$ BERT for bats (multiset dominates); BERT $\gg$ bag for marmosets.

Length is a confound. Fig. 2: order is n.s. at short sequences in both species; effects appear only at length, and at the matched 41–96 frame band both species are n.s. We therefore report a small, length-modulated species contrast, not a clean per-band dissociation.

Mechanism: sequential dependency, not diversity. We refute a token-diversity explanation (bats are *more* diverse: 10.2 vs. 3.4 unique tokens/voc) and confirm a sequential-dependency one (Fig. 3): the previous token removes 77.5% of next-token uncertainty for marmosets vs. 13–33% for bats; the bat’s modest frame-level dependency is context-irrelevant acoustic smoothness.

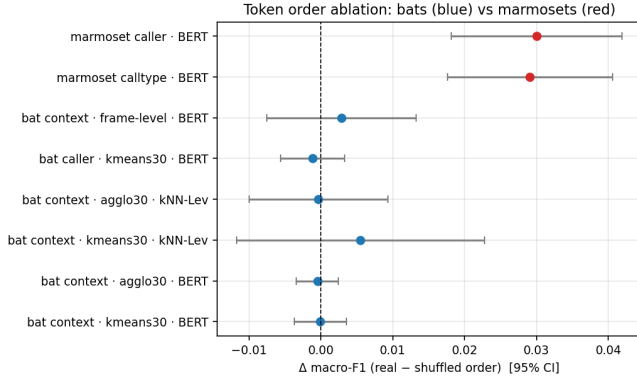


Figure 1: Order ablation. Bats (blue) straddle 0 (n.s.); marmosets (red) significant but small.

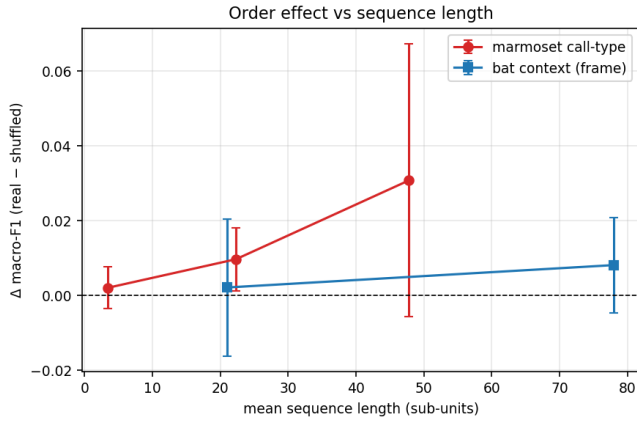


Figure 2: Order effect vs. sequence length.

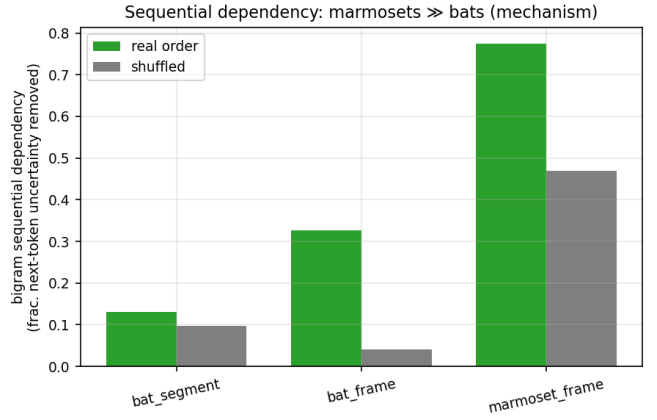


Figure 3: Bigram sequential dependency: marmosets $\gg$ bats.

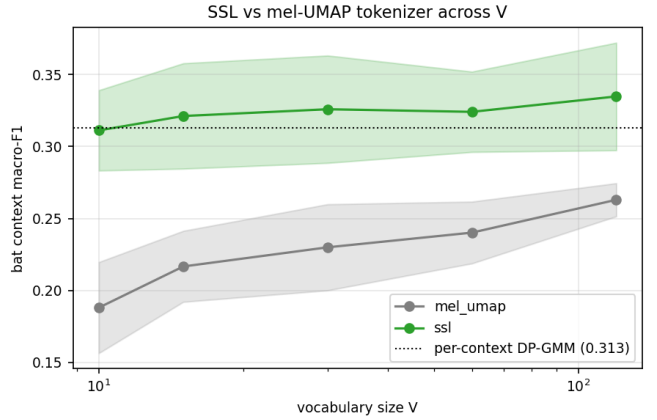


Figure 4: SSL vs. mel-UMAP tokenizer across vocabulary size V .

A domain-matched SSL tokenizer is the best discrete repertoire. Fig. 4: SSL k -means beats mel-UMAP at every V (bat context macro-F1 0.34 vs. 0.28), surpassing a per-context DP-GMM density baseline (0.31); discretization is near-lossless (+0.025 vs. continuous). SSL tokens also agree more with an independent DTW acoustic reference *despite lower silhouette* — silhouette mis-ranks quality.

The SSL advantage is task-dependent — and we show the cause. Fig. 5: varying the contrastive positive-pair definition causally shifts which task benefits. Same-vocalization positives maximize caller and destroy call-type; same-frame augmentation recovers call-type and loses caller. The invariance selects the served task; no SSL variant beats mel for fine within-call structure (call-type).

Leakage caveat. Per-context tokenization assigns disjoint token-ID ranges per context; a token-sequence classifier reads the label off the range (bag-LR macro-F1 ≈ 1.0). Per-context tokenizers must be evaluated by generative max-likelihood.

4 Discussion

Token-order effects in tokenized animal vocalizations are small and length-confounded; the multiset dominates. “Leverage sequential structure” claims need shuffle, length, and CI controls. A domain-matched SSL tokenizer is a strong, near-lossless discrete representation whose bias is set by the contrastive invariance. **Limitations:** small effect sizes limit per-band power; two species; analogous-but-not-identical featurization; closed-set marmoset caller-ID.

References

- [1] E. Sarkar and M. Magimai-Doss. Towards Leveraging Sequential Structure in Animal Vocalizations. arXiv:2511.10190, NeurIPS Workshop on AI for Non-Human Animal Communication, 2025.
- [2] Y. Prat et al. Everyday bat vocalizations contain information about emitter, addressee, context, and behavior. Sci. Rep., 2017.

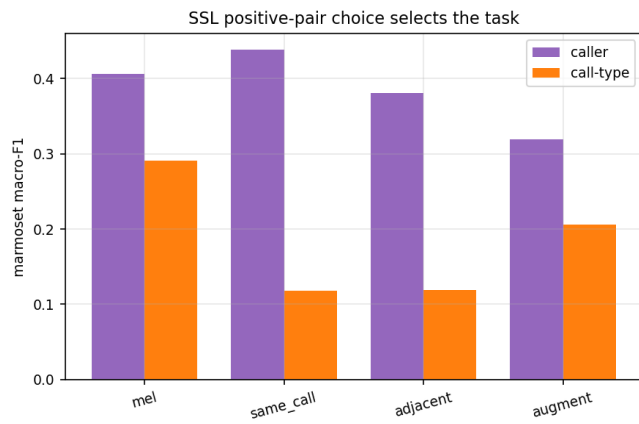


Figure 5: SSL positive-pair choice selects the downstream task.

- [3] M. Golyadkin, V. Pozdnyakov, L. Zhukov, I. Makarov. SensorSCAN: Self-supervised learning and deep clustering for fault diagnosis. Artif. Intell., 2023.